

DANA LAVACOT, PH.D.

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SUMMARY

Passionate and motivated **computational scientist** specializing in physics-based scientific machine learning (ML) and computational modeling of complex fluid phenomena. Possesses 5+ years of research experience in traditional scientific computing and developing scientific software with first author publications in high-impact journals. Impactful collaborator, experienced with diverse teams.

EDUCATION

Stanford University, Stanford, CA

September 2019 - August 2025

Ph.D in Mechanical Engineering

GPA: 4.06/4.0

Stanford University, Stanford, CA

September 2019 - June 2021

M.S. in Mechanical Engineering

GPA: 4.05/4.0

UC Berkeley, Berkeley, CA

August 2015 - May 2019

B.S. in Mechanical Engineering, High Honors

GPA: 3.875/4.0

RESEARCH EXPERIENCE

WASHINGTON UNIVERSITY IN ST. LOUIS, St. Louis, MO

August 2025 - Present

Postdoctoral Research Associate (PI: Fanwei Kong)

- Developed physics-informed diffusion-based model using Pytorch and CUDA for spatio-temporal cardiovascular flow reconstruction from heart ultrasound data
- Generated and processed large ML training dataset of cardiac blood flow using high-fidelity computational fluid dynamics simulations (100+ simulations of 10,000+ cells each) in SimVascular (C++/Fortran)
- Communicated complex technical concepts to clinical collaborators, enabling identification of needs and development of solution strategies
- Authored proposals for novel computational science and ML research, from conception to submission
- Mentored Ph.D. and undergraduate students on research projects in ML and computational modeling

STANFORD UNIVERSITY, Stanford, CA

January 2020 - August 2025

Graduate Researcher (PI: Ali Mani)

- Conducted 1,000+ high-fidelity hydrodynamic instability simulations using Lawrence Livermore National Lab's (LLNL) Ares (C/C++) and Pyrandra (Python/Fortran) codes on a high-performance computing cluster
- Performed simulation data analysis using novel data-driven method to develop a new turbulent mixing model, now integrated into LLNL's code for nuclear fusion experiment design
- Employed collaborative coding practices and version control using Git in adapting a parallel pseudo-spectral code (C++) for turbulence simulations
- Presented work in four first-author papers in high-impact journals and at research conferences

LAWRENCE LIVERMORE NATIONAL LABORATORY, Livermore, CA

June 2024 - September 2024

Defense Sciences and Technology Intern (PI: Brandon Morgan)

- Applied numerical method developed during my Ph.D. to assess turbulence models for hydrodynamic instability in nuclear fusion experiments, identifying a critical need for density ratio adjustments
- Developed scientific software in Python to numerically solve turbulence models and perform assessment
- Effectively communicated complex research in presentations to LLNL scientists and DSTI scholars across disciplines

UNIVERSITY OF CALIFORNIA, BERKELEY, Berkeley, CA

August 2017 - May 2019

Undergraduate Researcher (PI: Philip Marcus)

- Derived and programmed analytical gradients for a novel neural network layer designed for unstructured grids
- Built and trained an ML model using Pytorch and CUDA for an airfoil shape optimization task to demonstrate the effectiveness of the layer
- Co-authored publication presented at top computer vision conference (ICCV)

AWARDS AND HONORS

- **Stanford Graduate Fellowship in Science & Engineering**, 3-year award for top incoming Ph.D.s (2019)
- **UC Berkeley Dean's List**, semesterly distinction for GPAs in top 10% (2015-2018)
- **Boeing Scholarship**, awarded to outstanding STEM undergraduates (2016)
- **Banatao Scholarship**, awarded to 4 outstanding Filipino-American students in STEM (2015)

RESEARCH PAPERS

Lavacot, D. L. O.-L., Morgan, B. E., and Mani, A. (2026). Development and assessment of models for turbulent Rayleigh-Taylor mixing using the macroscopic forcing method. *Physica D: Nonlinear Phenomena*. <https://doi.org/10.1016/j.physd.2026.135136>.

Lavacot, D. L. O.-L., Mani, A., and Morgan, B. E. (2025). Atwood effects on nonlocality of the mean scalar transport operator in Rayleigh-Taylor mixing. *Physical Review Fluids*. <https://doi.org/10.1103/svjh-8pzl>.

Lavacot, D. L. O.-L., Liu, J., Morgan, B. E., and Mani, A. (2025). Techniques for improved statistical convergence in quantification of eddy diffusivity moments. *Physical Review Fluids*. <https://doi.org/10.1103/yrnt-y4mp>.

Lavacot, D. L. O.-L., Liu, J., Williams, H., Morgan, B. E., and Mani, A. (2024). Assessment of nonlocality of mean scalar transport in Rayleigh-Taylor instability using the Macroscopic Forcing Method. *Journal of Fluid Mechanics*. <https://doi.org/10.1017/jfm.2024.323>.

Jiang, C., **Lansigan, D. L. O.**, Marcus, P., and Niessner, M. (2019). DDSL: Deep Differentiable Simplex Layer for learning geometric signals. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.

CONFERENCE PRESENTATIONS

Lavacot, D. L. O.-L. and Kong, F. (2026). Uncertainty-aware cardiovascular flow reconstruction using a latent diffusion model. To be presented at the 10th World Congress of Biomechanics, Vancouver, Canada.

Lavacot, D. L. O.-L., Morgan, B. E., and Mani, A. (2024). Atwood effects on nonlocality of mean scalar transport in 3D Rayleigh-Taylor Instability. Presented at the APS Division of Fluid Dynamics 77th Annual Meeting, Salt Lake City, UT.

Lavacot, D. L. O.-L., Liu, J., Morgan, B. E., and Mani, A. (2023). Assessment of RANS models for Rayleigh-Taylor mixing using the Macroscopic Forcing Method. Presented at the APS Division of Fluid Dynamics 76th Annual Meeting, Washington, D.C.

Lavacot, D. L. O.-L., Liu, J., Morgan, B. E., and Mani, A. (2022). Continuing Investigations of Nonlocality in Rayleigh-Taylor Instability Using the Macroscopic Forcing Method. Presented at the APS Division of Fluid Dynamics 75th Annual Meeting, Indianapolis, IN.

Lansigan, D. L. O., Liu, J., Williams, H., Morgan, B. E., and Mani, A. (2021). Evaluating the Importance of Nonlocal Eddy Diffusivity for Rayleigh Taylor Instability. Presented at the APS Division of Fluid Dynamics 74th Annual Meeting, Phoenix, AZ.

Lansigan, D. L. O., D. Park, and Mani, A. (2020). An Accelerated Macroscopic Forcing Method for Determining Eddy Viscosity Operators. Presented at the APS Division of Fluid Dynamics 73rd Annual Meeting, Chicago, IL.

Lansigan, D. L. O., Jiang, C., and Marcus, P. (2018). Neural Network Powered Adjoint Methods: Gradient Based Shape Optimization with Deep Learning. Presented at the APS Division of Fluid Dynamics 71st Annual Meeting, Atlanta, GA.

INDUSTRY EXPERIENCE

GENERAL ATOMICS, ASI, Poway, CA

June 2023 - August 2023

Aero/CFD/HPC Tools Intern

- Assessed capabilities of STAR-CCM+ solver through 2D & 3D RANS simulations of airfoils and full fixed-wing aircraft, as part of evaluation of the CFD software presented to Engineering VP
- Demonstrated software capabilities with a simulation of flow over the MQ-9B aircraft, the largest simulation of the study (180M+ cells)
- Prepared documentation of best-practice simulation procedures to facilitate team's efficient transition to STAR-CCM+

TEACHING

WASHINGTON UNIVERSITY IN ST. LOUIS - *Guest Lecturer*

- Presented guest lecture on diffusion models to class of ~20 graduate students

STANFORD UNIVERSITY- *Graduate Teaching Assistant*

- Assisted in teaching Turbulence (Spring 2023, 20 students, graduate) and Numerical Methods (Spring 2022, 20 students, graduate), and undergraduate course, Vector Calculus (Fall 2024, 140 students, undergraduate)
- Supported instruction by facilitating office hours, designing exams, and evaluating assignments
- Presented guest lecture on Green's Theorem for Vector Calculus

UNIVERSITY OF CALIFORNIA- *Undergraduate Student Instructor*

- Taught weekly mini-lectures for Intro to Circuits & Linear Algebra (Fall 2018 & Spring 2019, ~1,000 students, undergraduate) in two discussion sections with ~40 students each